A PERFECT OBJECT ON A SCHEME WITH THE RESOLUTION PROPERTY IS REPRESENTED BY A BOUNDED COMPLEX OF FINITE LOCALLY FREE MODULES

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Allowed standard prerequisites (editorial).

- A perfect complex on a quasi-compact scheme has bounded tor amplitude and its cohomology modules are of finite type. invoked at proof lines 9800-9808 and again at proof lines 9820-9825. Provides the interval on which the induction runs and the finiteness used to build the surjection in the induction step.
- A finitely presented module of tor dimension zero is finite locally free. invoked at proof lines 9812-9818. Identifies the top cohomology in the base case with a finite locally free module.
- On a scheme with the resolution property the surjection onto a finite type module can be realized by a finite locally free module, and the kernel of such a surjection between finite locally free modules sits in an exact sequence controlled by the cone. invoked at proof lines 9825-9887. Produces the finite locally free module and the distinguished triangle used to reduce the interval.

Source reference. Original lemma 08CQ: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 9800-9802 to obtain the integers a and b bounding the tor amplitude; the induction set up on that interval is inside the counted span.

Source reference. Original lemma 08DN: sources/stacks-chow/tags.txt, source lines 1–1. Cited twice, at proof lines 9812-9816 for the base case and at proof lines 9820-9825 for the finiteness of the top cohomology in the induction step; both uses are inside the counted span.

Source reference. Original lemma 05P2: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 9816-9818 to conclude that the top cohomology in the base case is finite locally free; the base case argument is inside the counted span.

Source reference. Original lemma 01PG: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 9826-9840 to realize the surjection onto the top cohomology

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by a finite locally free module, which is where the resolution property enters; the triangle built from it is inside the counted span.

Source reference. Original definition 014E: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 9851-9855 for the distinguished triangle whose cone has a shorter tor amplitude interval; the induction that uses it is inside the counted span.

Source reference. Original lemma 08CJ: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 9856-9860 to see that the cone again has bounded tor amplitude, with a shorter interval; the application of the induction hypothesis to it is inside the counted span.

Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme with the resolution property. Let $\mathcal{F}^\bullet$ be a bounded below complex of quasi-coherent $\mathcal{O}_X$-modules representing a perfect object of $D(\mathcal{O}_X)$. Then there exists a bounded complex $\mathcal{E}^\bullet$ of finite locally free $\mathcal{O}_X$-modules and a quasi-isomorphism $\mathcal{E}^\bullet \to \mathcal{F}^\bullet$.*

Proof. Let $a, b \in \mathbf{Z}$ be integers such that $\mathcal{F}^\bullet$ has tor amplitude in $[a, b]$ and such that $\mathcal{F}^n = 0$ for $n < a$. The existence of such a pair of integers follows from Cohomology, Lemma 08CQ and the fact that X is quasi-compact. If $b < a$, then $\mathcal{F}^\bullet$ is zero in the derived category and the lemma holds. We will prove by induction on $b - a \geq 0$ that there exists a complex $\mathcal{E}^a \to \ldots \to \mathcal{E}^b$ with $\mathcal{E}^i$ finite locally free and a quasi-isomorphism $\mathcal{E}^\bullet \to \mathcal{F}^\bullet$.

The base case is the case $b - a = 0$. In this case $H^b(\mathcal{F}^\bullet) = H^a(\mathcal{F}^\bullet) = \mathrm{Ker}(\mathcal{F}^a \to \mathcal{F}^{a+1})$ is finite locally free. Namely, it is a finitely presented $\mathcal{O}_X$-module of tor dimension 0 and hence finite locally free. See Cohomology, Lemmas 08CQ and 08DN and Properties, Lemma 05P2. Thus we can take $\mathcal{E}^\bullet$ to be $H^b(\mathcal{F}^\bullet)$ sitting in degree b. The rest of the proof is dedicated to the induction step.

Assume $b > a$. Observe that

$$H^b(\mathcal{F}^\bullet) = \mathrm{Ker}(\mathcal{F}^b \to \mathcal{F}^{b+1}) / \mathrm{Im}(\mathcal{F}^{b-1} \to \mathcal{F}^b)$$

is a finite type quasi-coherent $\mathcal{O}_X$-module, see Cohomology, Lemmas 08CQ and 08DN. Then we can find a finite type quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ and a map

$$\mathcal{F} \longrightarrow \mathrm{Ker}(\mathcal{F}^b \to \mathcal{F}^{b+1})$$

such that the composition with the projection onto $H^b(\mathcal{F}^\bullet)$ is surjective. Namely, we can write $\mathrm{Ker}(\mathcal{F}^b \to \mathcal{F}^{b+1})$ as the filtered union of its finite type quasi-coherent submodules by Properties, Lemma 01PG and then one of these will do the job. Next, we choose a finite locally free $\mathcal{O}_X$-module $\mathcal{E}^b$ and a surjection $\mathcal{E}^b \to \mathcal{F}$ using the resolution property of X. Consider the map of complexes

$$\alpha : \mathcal{E}^b[-b] \to \mathcal{F}^\bullet$$

and its cone $C(\alpha)^\bullet$, see Derived Categories, Definition 014E. We observe that $C(\alpha)^\bullet$ is nonzero only in degrees $\geq a$, has tor amplitude in $[a, b]$ by Cohomology, Lemma 08CJ, and has $H^b(C(\alpha)^\bullet) = 0$ by construction. Thus we actually find that $C(\alpha)^\bullet$ has tor amplitude in $[a, b-1]$. Hence the induction hypothesis applies to $C(\alpha)^\bullet$ and we find a map of complexes

$$(\mathcal{E}^a \to \ldots \to \mathcal{E}^{b-1}) \longrightarrow C(\alpha)^\bullet$$

with properties as stated in the induction hypothesis. Unwinding the definition of the cone this gives a commutative diagram

$$\begin{array}{ccccccccc} \ldots & \longrightarrow & \mathcal{E}^{b-2} & \longrightarrow & \mathcal{E}^{b-1} & \longrightarrow & 0 & \longrightarrow & \ldots \\ & & \downarrow & & \downarrow & & \downarrow & & \\ \ldots & \longrightarrow & \mathcal{F}^{b-2} & \longrightarrow & \mathcal{F}^{b-1} \oplus \mathcal{E}^b & \longrightarrow & \mathcal{F}^b & \longrightarrow & \ldots \end{array}$$

It is clear that we obtain a map of complexes $(\mathcal{E}^a \to \ldots \to \mathcal{E}^b) \to \mathcal{F}^\bullet$. We omit the verification that this map is a quasi-isomorphism. $\square$